

Part I: Test Results and Analysis

After conducting the planned tests for our 3D-printed tree stand, we gathered and analyzed the following results:

Scratch Test:

-A total of 15 adult participants tested the tree stand, with 4 participants reporting scratches. One outlier reported 2 scratches during the test. Upon inspecting the tree stand, we identified the scratch points and documented them for improvement. Future iterations will include sanding and smoothing of the identified areas. Overall, we are pleased with the low number of scratches reported, indicating that our surface finish is mostly effective.

Drop Test:

-To carry out the drop test, we selected 6 individuals of varying strength levels and printed 6 tree stands, ensuring no stand was dropped more than once. Weak participants (able to lift under 50 lbs) did not manage to cause any breakage. Average participants (50-100 lbs) managed to fracture one of the stands at a stress point, while strong participants (100-200 lbs) managed to completely break one stand. This test demonstrated that while the tree stand is durable under typical use, it does have weak points that we plan to reinforce in future iterations.

Load Test:

-For the load test, we printed three additional tree stands and tested them with incremental weights and time durations. We were impressed with the structure's durability, as it withstood 15 lbs for 30 minutes without collapsing. Although our weight endurance is promising, improvements to internal support structures are planned to increase load capacity for more extreme outdoor conditions.

Setup Test:

-We tested three different instructional methods for setting up the tree stand with 30 participants. Group A, which received only written instructions, had a low success rate. Group B, with written instructions and live demonstration, showed marked improvement. Group C, which utilized a video tutorial, had the highest success rate. We recommend using a combination of written and video guides for future implementations.

Overall, the tests demonstrated that our 3D-printed tree stand design is both robust and intuitive to use, though there are areas for improvement. We intend to address the weaknesses identified during testing to ensure maximum safety and reliability for outdoor use.